

IT3091 – Machine Learning

25 Code-Based Multi-Select MCQs

Lab Sheets 06–08: Regression, SVM/SVR, Ensembles and Clustering

Suggested duration: 65 minutes
Number of questions: 25
Options per question: Five (A–E)
Question type: Select all correct answers

Instructions

- Every question has exactly **two or three** correct options.
- Select every fully correct statement. Partial selections are not complete answers.
- Interpret each code fragment using standard NumPy and Scikit-learn behaviour.
- Assume all required imports and valid data objects have already been created unless stated otherwise.

01. Consider the prediction function below. Which statements are correct?

```
def predict(X, beta0, beta1):  
    return beta0 + beta1 * X
```

- ☐ **A.** For a NumPy vector `X`, the function returns one prediction per element.
- ☐ **B.** Increasing `beta0` by 2 adds 2 to every prediction.
- ☐ **C.** The function converts a continuous target into class labels.
- ☐ **D.** The multiplication changes `X` in place.
- ☐ **E.** The value `beta1` represents the residual of the first observation.

02. A student completes the MSE function as follows. Which interpretations are correct?

```
def mse(y, y_hat):  
    residuals = y - y_hat  
    return np.mean(residuals ** 2)
```

- ☐ **A.** The returned result is Mean Absolute Error.
- ☐ **B.** The returned result is Root Mean Squared Error.
- ☐ **C.** A positive residual means the fitted value is below the observed value.
- ☐ **D.** Squaring prevents positive and negative residuals from cancelling.
- ☐ **E.** The function returns the average squared residual as one scalar.

03. Which statements correctly describe this gradient-descent update?

```
db0 = (2/n) * np.sum(y_hat - y)  
db1 = (2/n) * np.sum((y_hat - y) * X)  
beta0 -= learning_rate * db0  
beta1 -= learning_rate * db1
```

- ☐ **A.** Each coefficient moves opposite to its calculated gradient.
- ☐ **B.** The slope gradient is calculated without using the predictor values.
- ☐ **C.** Appending MSE after each update can be used to examine convergence.
- ☐ **D.** The learning rate automatically determines the number of iterations.
- ☐ **E.** Replacing `y_hat - y` by `y - y_hat` without changing the update sign gives the same descent step.

04. Examine the Scikit-learn verification code. Which statements are correct?

```
X2 = X.reshape(-1, 1)
model = LinearRegression().fit(X2, y)
```

- ☐ A. The reshape creates one row containing all predictor values.
- ☐ B. The fitted input matrix has one column and one row per observation.
- ☐ C. The slope is stored in `model.intercept_`.
- ☐ D. `model.predict([[4.5]])` is a valid one-observation prediction call.
- ☐ E. `model.intercept_` and `model.coef_[0]` can be compared with the from-scratch estimates.

05. After fitting a linear SVC, the following values are extracted. Which statements are correct?

```
w = svc.coef_[0]
b = svc.intercept_[0]
scores = X @ w + b
```

- ☐ A. The manual scores should agree with `svc.decision_function(X)`.
- ☐ B. Every value in `scores` is automatically a calibrated probability.
- ☐ C. A negative score is assigned to the positive class in the lab's binary setup.
- ☐ D. `w` is a coefficient vector and `b` is a scalar intercept.
- ☐ E. `svc.support_` contains the coordinate vectors of the support points.

06. Which statements correctly interpret the vectorized hinge-loss calculation?

```
scores = svc.decision_function(X)
hinge = np.maximum(0, 1 - y * scores)
```

- ☐ A. It computes one hinge-loss value for every observation without an explicit loop.
- ☐ B. The loss is zero whenever the signed margin `y * scores` is at least 1.
- ☐ C. A misclassified observation has positive hinge loss.
- ☐ D. Every support vector must have exactly zero hinge loss.
- ☐ E. `np.maximum` converts the scores into class probabilities.

07. A linear SVR is fitted using the code below. Which statements are correct?

```
Xr = X.reshape(-1, 1)
svr = SVR(kernel='linear', C=1.0, epsilon=1.0)
svr.fit(Xr, y)
```

- ☐ A. `svr.predict(Xr)` returns continuous fitted values.
- ☐ B. The fitted object can output only the class labels -1 and $+1$.
- ☐ C. The value of `epsilon` is ignored when the kernel is linear.
- ☐ D. The target `y` must contain exactly two distinct categories.
- ☐ E. For the linear kernel, coefficients and intercept can reconstruct the fitted predictions.

08. The following SVR quantities are calculated. Which statements are correct?

```
preds = svr.predict(Xr)
upper = preds + svr.epsilon
lower = preds - svr.epsilon
inside = np.abs(y - preds) < svr.epsilon
loss = np.maximum(0, np.abs(y - preds) - svr.epsilon)
```

- ☐ A. `inside` includes observations exactly on the tube boundary.
- ☐ B. `upper` and `lower` define the two sides of the epsilon tube.
- ☐ C. A True value in `inside` implies zero epsilon-insensitive loss.
- ☐ D. `loss` is simply the absolute error for every observation.
- ☐ E. `loss` penalizes only the part of absolute error extending beyond the tube.

09. Which statements about this scaling pattern are correct?

```
scaler = StandardScaler()
Xtr = scaler.fit_transform(X_train)
Xte = scaler.transform(X_test)
```

- ☐ A. The scaler is fitted independently a second time on the test set.
- ☐ B. Training statistics are used for both the training and test transformations.
- ☐ C. The pattern avoids using test-set means and standard deviations during fitting.
- ☐ D. The code automatically standardizes `y_train`.
- ☐ E. This preprocessing guarantees perfect SVM or SVR performance.

10. Assume a binary target. Which statements correctly describe this split?

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42)
```

- ☐ A. Using the same data and seed makes the split reproducible.
- ☐ B. The model should be trained directly on `X_test`.
- ☐ C. `random_state=42` increases the number of observations.
- ☐ D. `stratify=y` aims to preserve class proportions across the two subsets.
- ☐ E. The function returns separate predictor and target objects for training and testing.

11. Which statements correctly interpret this Random Forest configuration?

```
rf = RandomForestClassifier(
    n_estimators=100, max_depth=4,
    max_features='sqrt', random_state=42)
```

- ☐ A. Each new tree is trained to correct the residuals of the preceding tree.
- ☐ B. The ensemble contains 100 trees.
- ☐ C. Each tree is restricted to exactly four training observations.
- ☐ D. Only a subset of predictors is considered when searching for each split.
- ☐ E. The fixed random state guarantees the best possible test accuracy.

12. A fitted Random Forest is summarized as follows. Which statements are correct?

```
imp = rf.feature_importances_  
order = np.argsort(imp)[::-1]  
top_names = feature_names[order[:10]]
```

- ☐ A. order places the largest importances first.
- ☐ B. The result proves that changing a top feature will causally change the target.
- ☐ C. top_names remains aligned with the selected importance positions.
- ☐ D. Negative values indicate that a feature decreases the predicted class.
- ☐ E. The values summarize split-based contribution within the fitted forest.

13. A grid search uses the configuration below. Which statements are correct?

```
grid = {'n_estimators':[50,100],  
       'learning_rate':[0.05,0.1],  
       'max_depth':[1,2]}  
search = GridSearchCV(model, grid, cv=5)
```

- ☐ A. The independent test set is used to choose every parameter combination.
- ☐ B. The grid contains $2 \times 2 \times 2 = 8$ candidate combinations.
- ☐ C. best_params_ stores the fitted coefficients of the winning model.
- ☐ D. The value cv=5 means that five predictor columns are required.
- ☐ E. The search performs 40 train-validation fits, excluding any final refit.

14. Which statements correctly interpret this Gradient Boosting model?

```
gb = GradientBoostingClassifier(  
    n_estimators=100, learning_rate=0.05,  
    max_depth=2)
```

- ☐ A. The model adds learners sequentially.
- ☐ B. All trees are independent bootstrap models that can be trained in parallel.
- ☐ C. The learning rate shrinks the contribution of each newly added learner.
- ☐ D. max_depth controls the complexity of the individual trees.
- ☐ E. Increasing all three values is guaranteed to improve test accuracy.

15. Which statements are correct for the evaluation code below?

```
pred = model.predict(X_test)  
acc = accuracy_score(y_test, pred)
```

- ☐ A. pred must contain calibrated probabilities.
- ☐ B. acc measures performance on the training subset.
- ☐ C. y_test and pred must contain corresponding outcomes of equal length.
- ☐ D. The calculation is appropriate for classification labels.
- ☐ E. The result is the average continuous regression residual.

16. Study the data-generation and preprocessing code. Which statements are correct?

```
X, hidden = make_blobs(n_samples=180, centers=3,
                        cluster_std=1.15, random_state=42)
X_scaled = StandardScaler().fit_transform(X)
```

- ☐ A. The variable `hidden` should not be supplied as the target to the clustering algorithm.
- ☐ B. `X_scaled` has the same number of rows and columns as `X`.
- ☐ C. The code changes the two feature columns into three feature columns.
- ☐ D. Clustering is trained by calling `fit(X_scaled, hidden)`.
- ☐ E. Both feature columns are standardized using statistics calculated from `X`.

17. Which statements correctly describe this centroid initialization?

```
rng = np.random.default_rng(42)
idx = rng.choice(len(X_scaled), size=K, replace=False)
centroids = X_scaled[idx]
```

- ☐ A. The same observation may be selected several times in one initialization.
- ☐ B. The array `idx` contains final cluster labels for all observations.
- ☐ C. The initial centroids are K distinct observations from the scaled data.
- ☐ D. The code has already completed every K-Means assignment and update step.
- ☐ E. The fixed seed allows the same initialization to be reproduced.

18. A K-Means iteration is implemented as follows. Which statements are correct?

```
dist = np.linalg.norm(
    X_scaled[:, None, :] - centroids[None, :, :], axis=2)
labels = np.argmin(dist, axis=1)
new_centroids = np.array([
    X_scaled[labels == k].mean(axis=0) for k in range(K)])
```

- ☐ A. `dist` contains only one distance for the complete dataset.
- ☐ B. `dist` has one row per observation and one column per centroid.
- ☐ C. `argmin(..., axis=1)` assigns each observation to its nearest centroid.
- ☐ D. The list comprehension updates each centroid using its assigned observations.
- ☐ E. The update uses coordinate-wise medians rather than means.

19. A final from-scratch clustering is evaluated using the code below. Which statements are correct?

```
wcss = np.sum((X_scaled - centroids[labels]) ** 2)
km = KMeans(n_clusters=K, n_init=10, random_state=42)
km.fit(X_scaled)
```

- ☐ A. The first line sums unsquared Euclidean distances.
- ☐ B. K-Means attempts to maximize `wcss`.
- ☐ C. The numeric values in `labels` must equal `km.labels_` entry by entry.
- ☐ D. The first line accumulates squared within-cluster deviations.
- ☐ E. `km.inertia_` is the Scikit-learn quantity comparable with the calculated WCSS.

20. Which statements correctly describe this Scikit-learn K-Means code?

```
km = KMeans(n_clusters=3, n_init=10, random_state=42)
labels = km.fit_predict(X_scaled)
```

- ☐ A. labels contains one cluster ID for every observation.
- ☐ B. km.cluster_centers_ has three rows.
- ☐ C. The cluster IDs are guaranteed to match the hidden make_blobs labels numerically.
- ☐ D. km.inertia_ stores the fitted within-cluster sum of squares.
- ☐ E. n_init=10 means that only one centroid initialization is considered.

21. Which statements correctly interpret the hierarchical-clustering code?

```
Z = linkage(X_scaled, method='complete')
dendrogram(Z)
```

- ☐ A. Z records the merge sequence and fusion distances.
- ☐ B. Complete linkage uses the farthest cross-cluster pair when measuring separation.
- ☐ C. The function requires known class labels as its second argument.
- ☐ D. The final number of clusters must be supplied before Z is constructed.
- ☐ E. The dendrogram axes must be identical to those of the original scatter plot.

22. Consider this agglomerative-clustering code. Which statements are correct?

```
agg = AgglomerativeClustering(
    n_clusters=3, linkage='average')
labels = agg.fit_predict(X_scaled)
```

- ☐ A. The fitted centroids are retrieved from agg.cluster_centers_.
- ☐ B. labels has one cluster assignment per observation.
- ☐ C. Average linkage uses average cross-cluster pairwise separation.
- ☐ D. The method trains from known target labels hidden inside X_scaled.
- ☐ E. Observations classified as noise automatically receive label -1.

23. Which statements correctly interpret this DBSCAN result summary?

```
labels = DBSCAN(eps=0.25, min_samples=5).fit_predict(X_moons)
n_clusters = len(set(labels)) - (1 if -1 in labels else 0)
n_noise = np.sum(labels == -1)
```

- ☐ A. The value -1 is included as an ordinary cluster in n_clusters.
- ☐ B. The maximum numeric label always equals the number of clusters.
- ☐ C. The expression for n_clusters excludes DBSCAN's noise label.
- ☐ D. n_noise counts observations marked as noise.
- ☐ E. The array labels contains class probabilities.

24. Suppose both models below are applied to the standardized moons data. Which statements are correct?

```
db_labels = DBSCAN(eps=0.25, min_samples=5).fit_predict(X_moons)
km_labels = KMeans(n_clusters=2, n_init=10,
                  random_state=42).fit_predict(X_moons)
```

- ☐ A. K-Means explicitly assigns isolated observations the permanent noise label -1 .
- ☐ B. DBSCAN is forced by this code to return exactly two clusters.
- ☐ C. Standardization guarantees that both methods produce identical assignments.
- ☐ D. DBSCAN may mark some moon observations as noise.
- ☐ E. Plotting the two label arrays side by side reveals their different clustering assumptions.

25. Which statements correctly describe this complete evaluation pattern?

```
scaler = StandardScaler().fit(X_train)
Xtr = scaler.transform(X_train)
Xte = scaler.transform(X_test)
model.fit(Xtr, y_train)
result = model.score(Xte, y_test)
```

- ☐ A. All scaling estimates are learned from the training predictors.
- ☐ B. The scaler should be refitted on `X_test` before transforming it.
- ☐ C. `model.score` always represents classification accuracy for every estimator.
- ☐ D. Using the same random seed guarantees strong generalization to future data.
- ☐ E. The test subset is used after model fitting to estimate performance on unseen cases.